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CONTROL DEVICE FOR DETERMINING A
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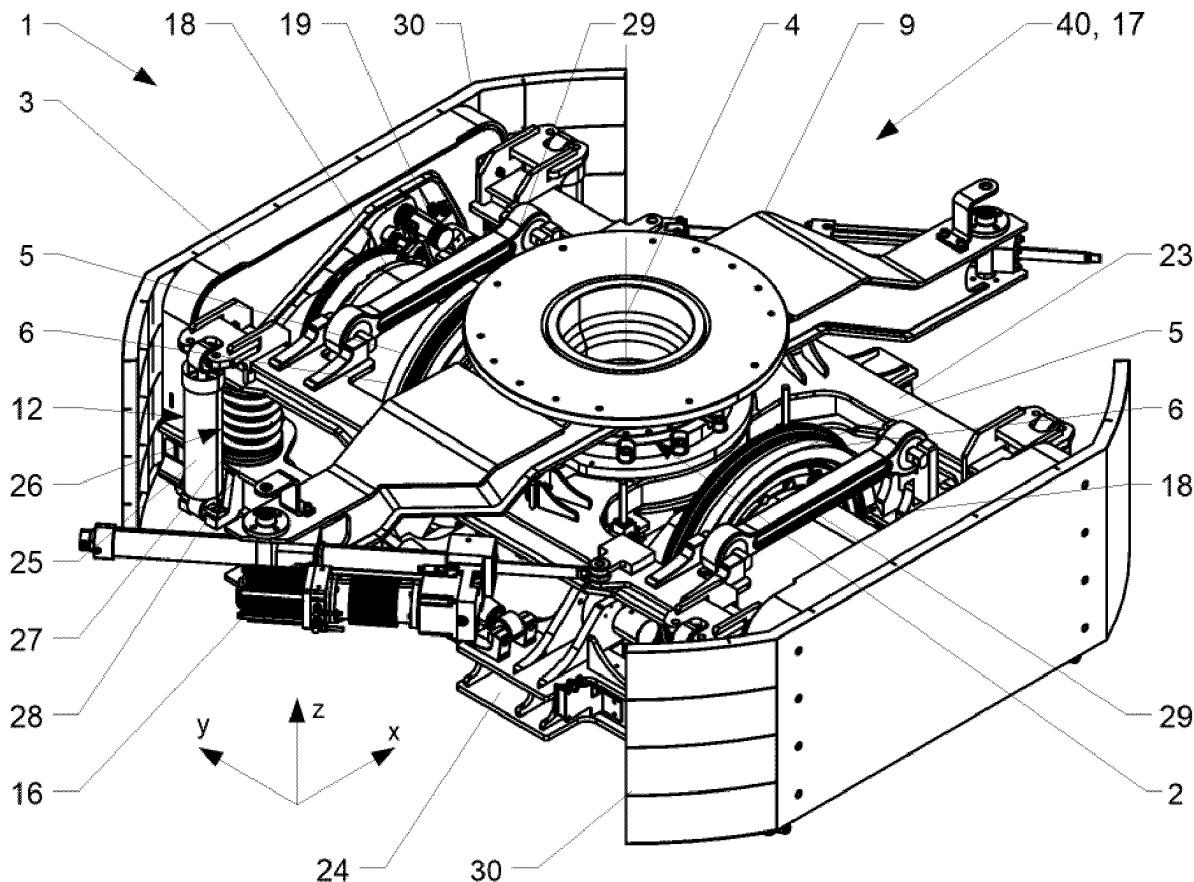
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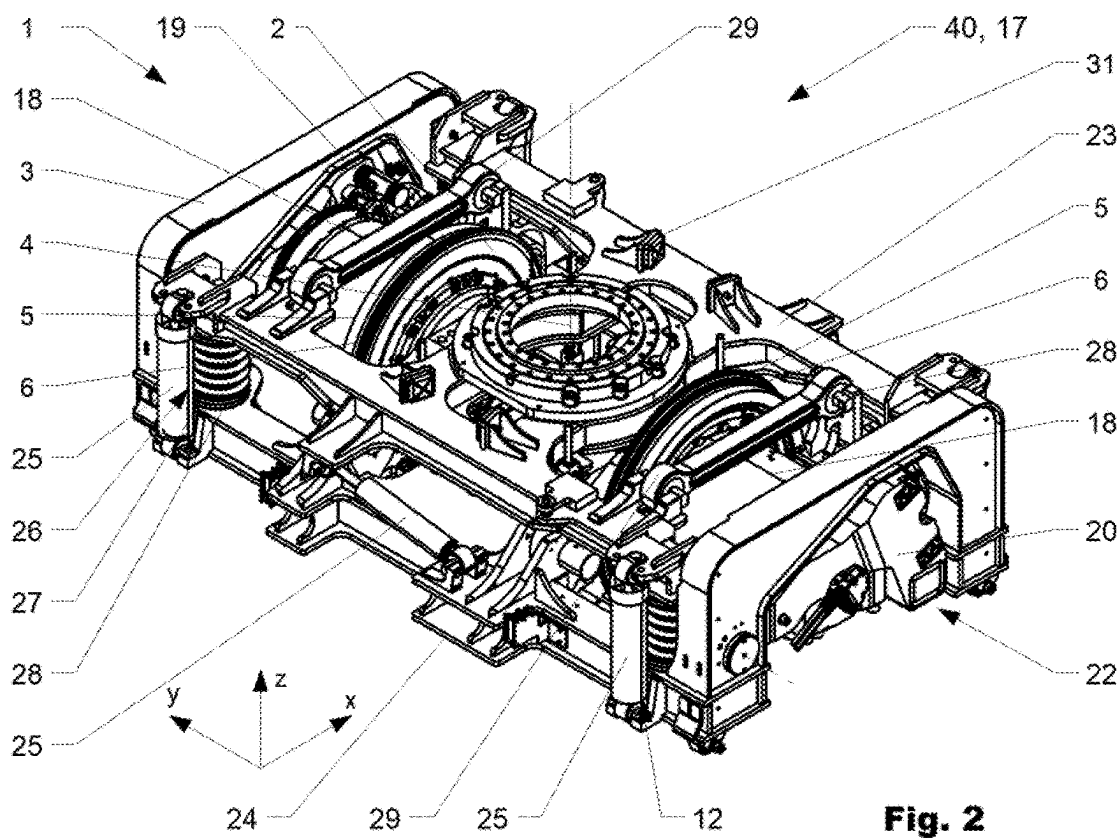
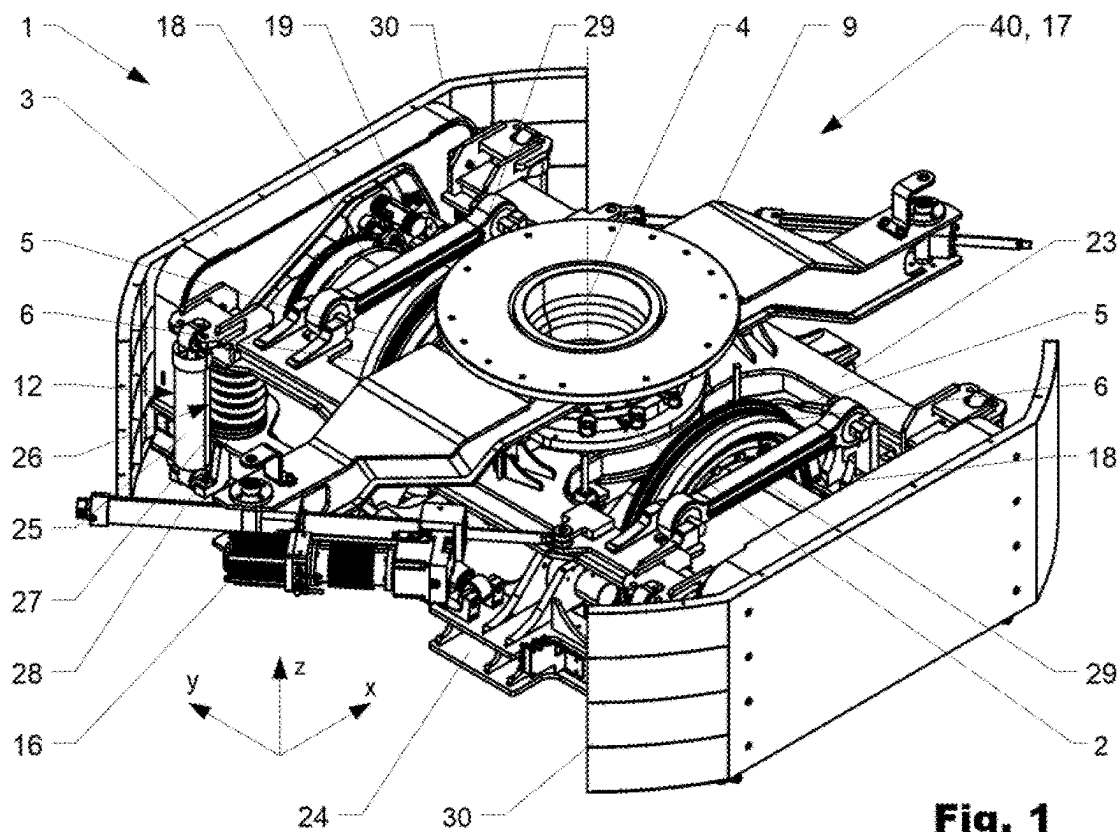
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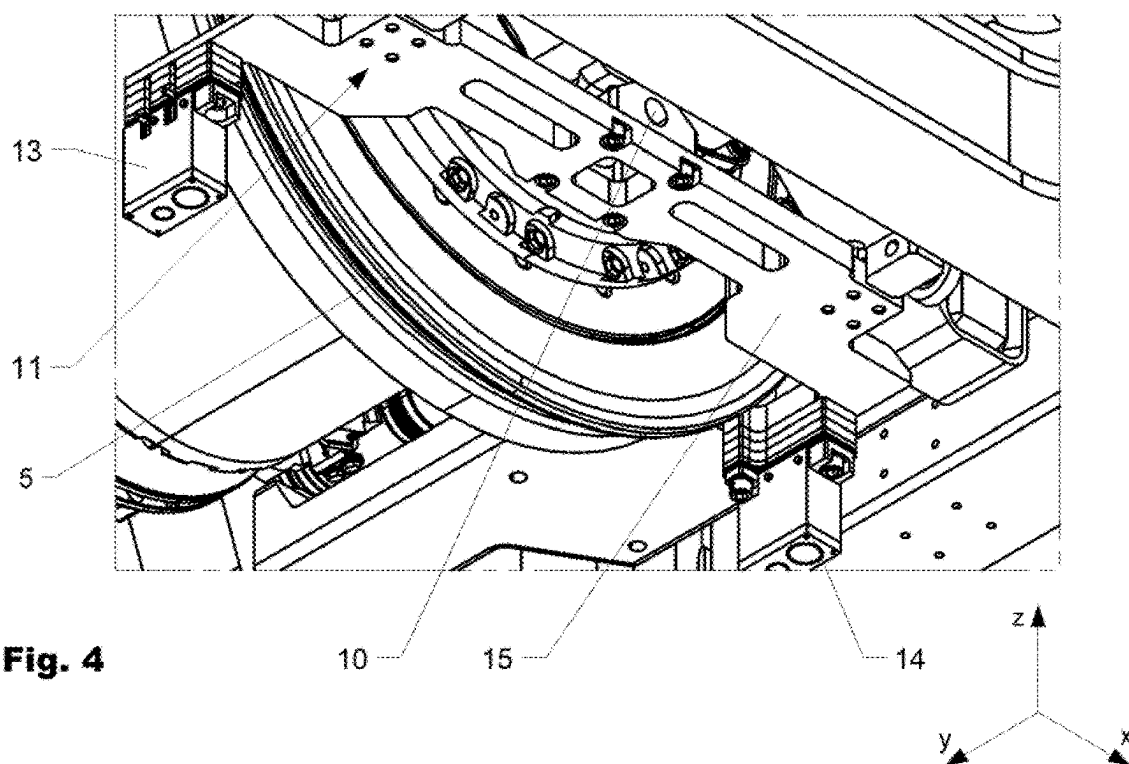
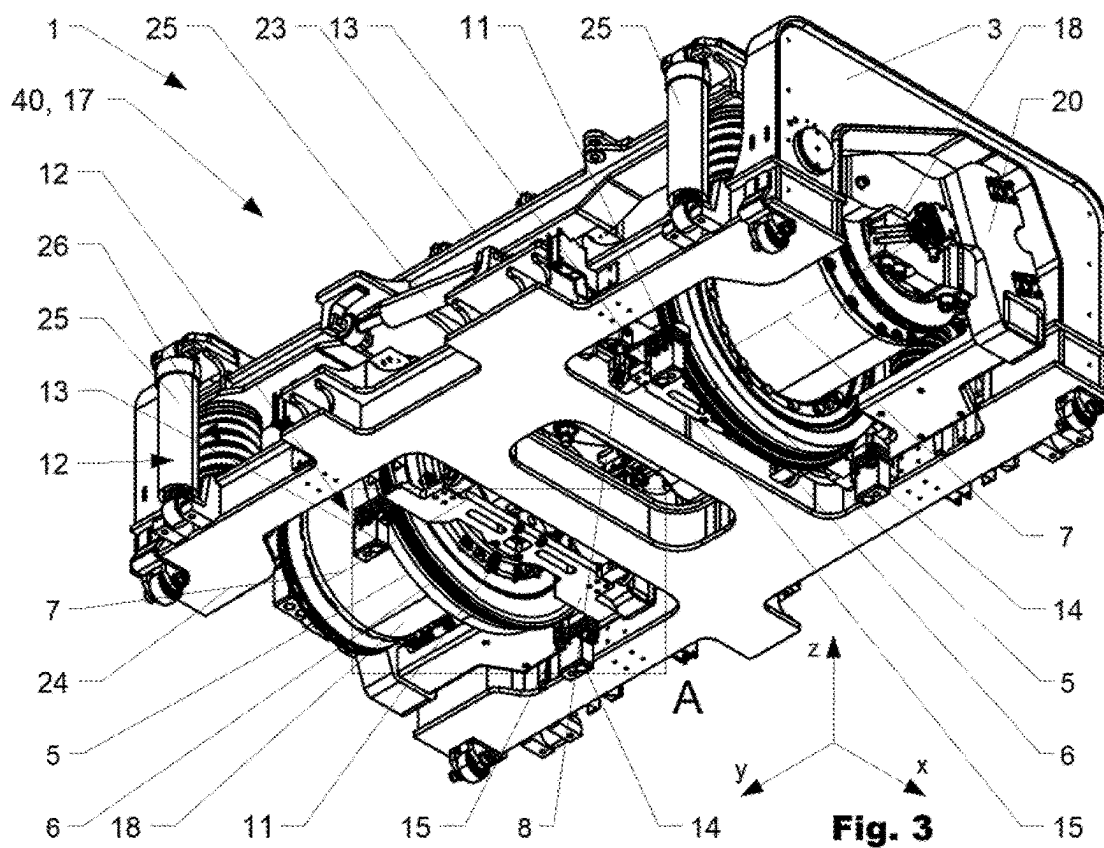
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ABSTRACT

The present disclosure relates to a computer-implemented method for determining a real time steering angle (58) for a railway bogie (1) and to a control device (40) which is configured to perform the method, the method comprising: deter-mining by a sensor assembly at least one lateral real time sensor signal (38), which is characteristic for a lateral position of a tread of at least one wheel of the railway bogie (1) with respect to a railway track, determining, by means of a positioning algorithm (47), a real time position and/or orientation of the railway bogie (1) with respect to the railway track (31), using the received at least one lateral real time sensor signal (38) and determining the real time steering angle (58) for steering of the railway bogie (1), using the determined real time position and/or orientation of the railway bogie (1).







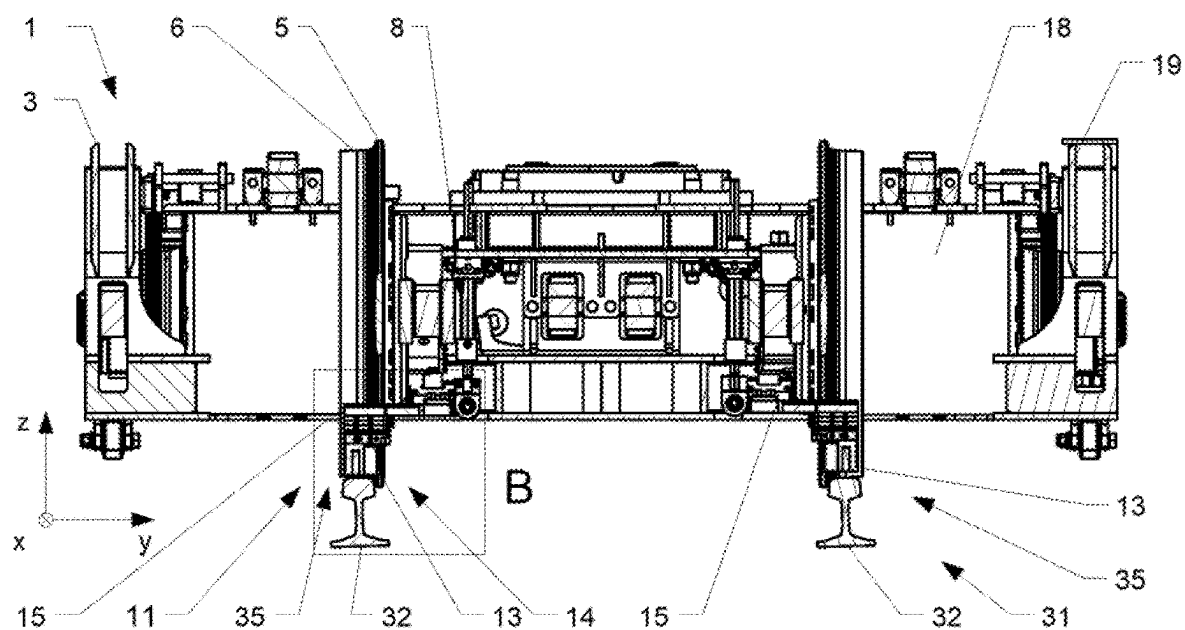


Fig. 5

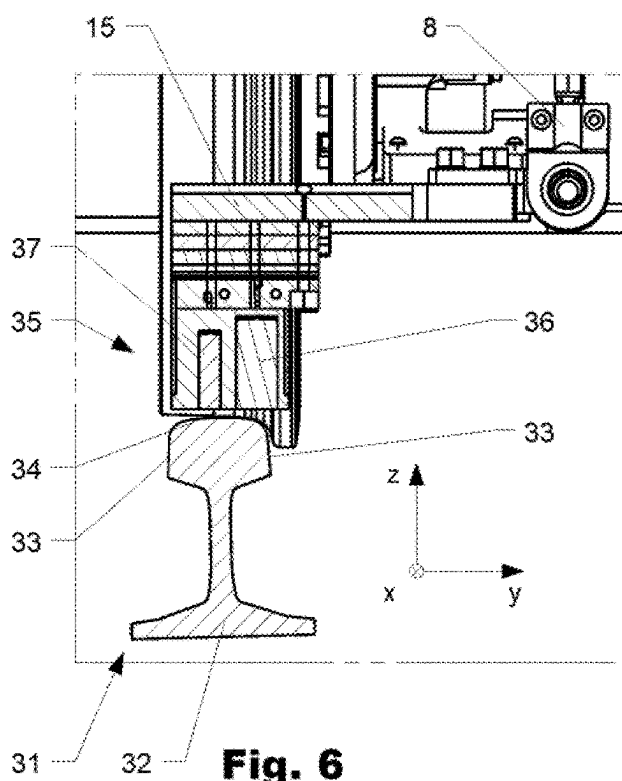


Fig. 6

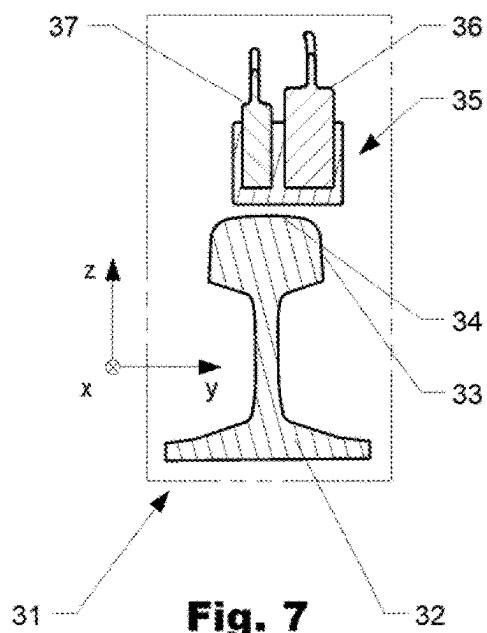
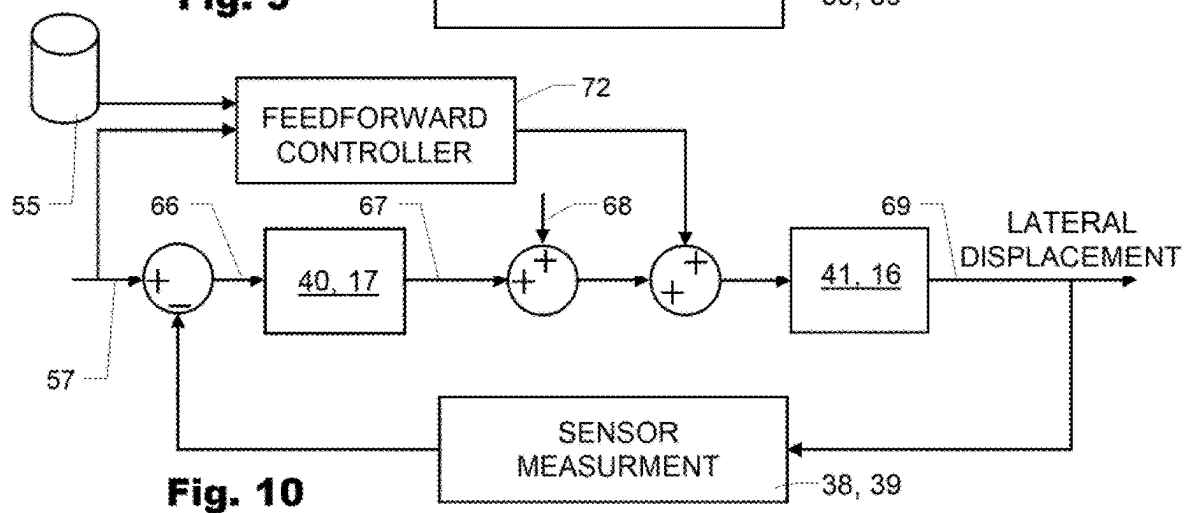
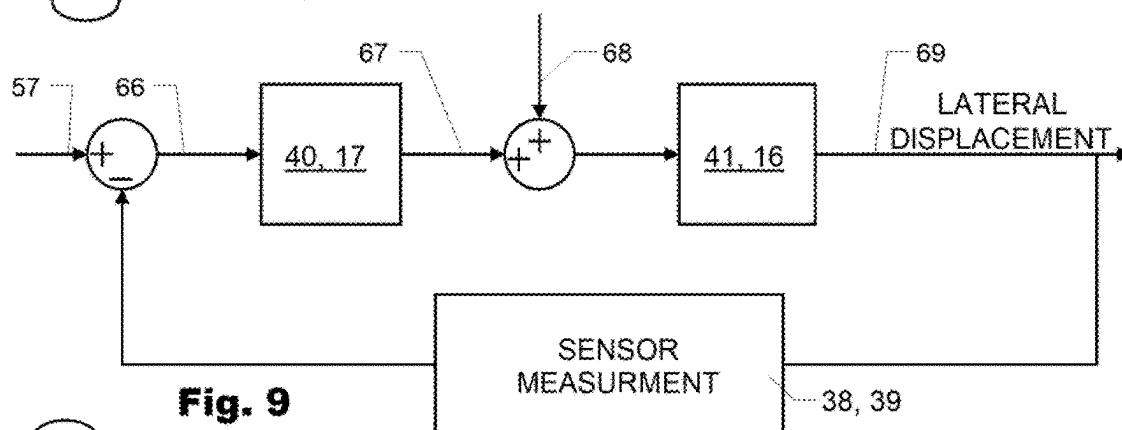
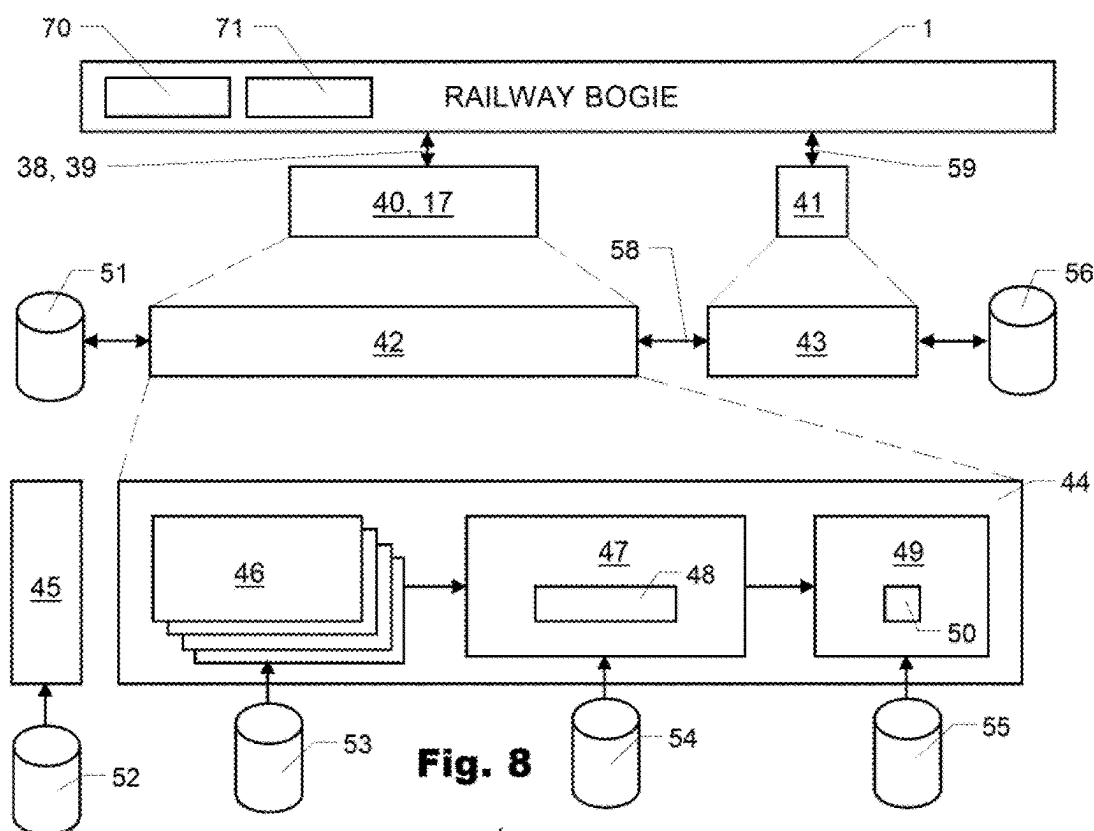


Fig. 7



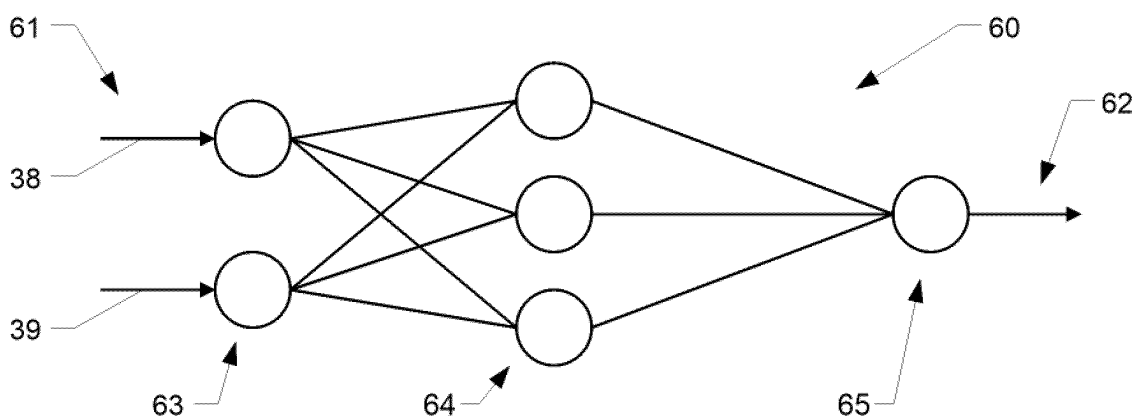


Fig. 11

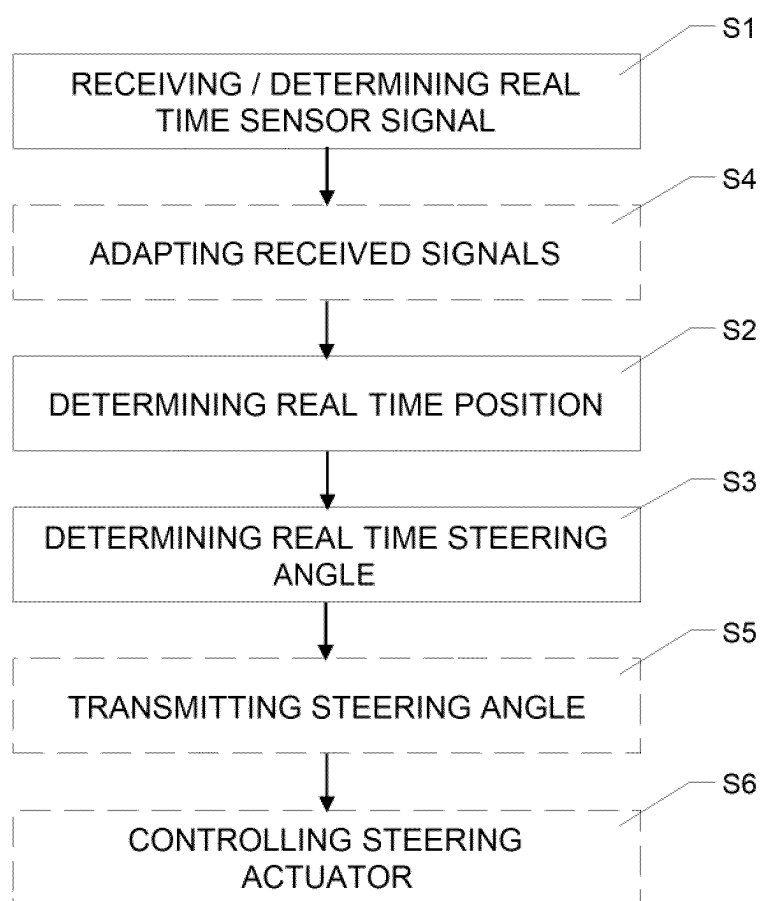


Fig. 12

COMPUTER-IMPLEMENTED METHOD AND CONTROL DEVICE FOR DETERMINING A REAL TIME STEERING ANGLE

FIELD OF THE DISCLOSURE

[0001] The present disclosure relates to a computer-implemented method and to a control device for determining a real time steering angle of a railway bogie. In particular, the present disclosure relates to a computer-implemented method determining a real time steering angle for a railway bogie, the method comprising a plurality of steps. Further, the present disclosure relates a control device for determining a real time steering angle for a railway bogie, the control device comprising a processor, which is configured to perform a plurality of steps for determining the real time steering angle.

BACKGROUND OF THE DISCLOSURE

[0002] Railway vehicles, which are track bound such as trains, trams (streetcars, tramway) or other vehicles often exhibit wheels that are not optimally aligned to the tracks leading to higher friction between the railway track and treads of the wheels. Especially in curves with a small radius, this contact leads to an increased profile wear of both the wheel and the rail, known as Rail Contact Fatigue (RCF), and noise pollution. In case of low-floor vehicles, this effect is even more pronounced: The low-floor vehicles feature smaller and less wheels per vehicle in order to increase the passenger comfort and inner space of the vehicle by having a continuous low-floor structure. However, this further leads to enhanced loads per wheel and a more pronounced fatigue of the wheel's material causing smaller rifts or even larger material fractures.

[0003] Several attempts are known to reduce the track and wheel wear. In the 1990's, systems have been developed that were able to steer the wheels in curves. However, it turned out that these solutions often suffered from undesired side effects in straight track sections such that the wheels adhered one-sided with the tread on the track, leading to an enhanced wear and noise in straight track sections. Hence, after a few years, most of these concepts were discarded and conventional concepts combined with wheel-noise absorbers and advanced industrial lubricants were again pursued. A particular challenge was and still is to determine accurately and reliably during operation of the bogie the position and orientation of the bogie with respect to the railway track.

[0004] One example of a railway bogie, which addresses these disadvantages in a successful manner is the WO2018015290 published 2018 in the name of the same applicant. The disclosed vehicle comprises a wheel assembly interconnected to a chassis as well as a method for steering said vehicle. The wheel assembly comprises a cross-member having a first end to which a first hub is interconnected by a first steering joint and a second end to which a second hub is interconnected by a second steering joint. A first wheel is attached to the first hub rotatable around a first rotation axis and a second wheel is attached to the second hub rotatable around a second rotation axis.

[0005] Trams or streetcars, which are driving on railway tracks with narrow curves with relatively low speeds, produce lots of noise while driving, in particular in curves when the flanges of the wheels hit the railway track. Even steered trams or streetcars do not have such a precise and fast

steering to avoid each noise producing contact of the flange of the wheel on the railway track. Noise is also generated when there is a stick and slip effect between wheels, that are commonly connected by an axle shafts, are forced to rotate at difference speeds as a vehicle moves through curved track sections.

SUMMARY OF THE DISCLOSURE

[0006] The object of the present disclosure is to provide a computer-implemented method and a control device for determining a real time steering angle for a railway bogie. In particular, it is an object of the present disclosure to provide a computer implemented method for determining a real time steering angle used by a railway bogie and a control device of a railway bogie for determining a real time steering angle for a railway bogie.

[0007] According to the present disclosure, these objects are addressed by the features of the independent claims. In addition, advantageous embodiments follow from the dependent claims and the description.

[0008] According to the present disclosure, a computer-implemented method for determining a real time steering angle for a railway bogie is specified. The method may comprise:

[0009] determining by a sensor assembly at least one lateral real time sensor signal, which is characteristic for a lateral position of a tread of at least one wheel of the railway bogie with respect to a railway track. The lateral position of the tread of the wheel determines the lateral distance of the tread and therefore the current real time lateral distance of a flange of the wheel to a rail of the railway track, in particular during operation of the railway bogie on the railway track.

[0010] determining, by means of a positioning algorithm, a real time position and/or orientation of the railway bogie with respect to the railway track, using the received at least one lateral real time sensor signal. The positioning algorithm uses the received at least one lateral real time sensor signal for determining the position and/or orientation of the railway bogie on the railway track. The position is for example a lateral displacement of the wheel flange with respect to a target or set point lateral displacement value. In a variation, the real time position may be the lateral displacement of the geometrical center of the railway bogie with respect to the current center of the railway track. The current center of the railway track is for example the half of the distance between the two rails of the railway track on the section of the railway track on which the railway bogie is currently in operation. The orientation of the railway bogie is for example determined by the angle between a rotation axis of the wheel (or the wheels) and the main extension direction of the respective rail of the railway track. The target position or set-point position is for example when the geometrical center of the railway bogie lies above the current center of the railway track and the target orientation or set-point orientation is for example when the angle between the rotation axis of the wheel and the respective rail is 90°. In other words, the rotation axis extends perpendicular with respect to the main extension direction of the respective rail, or the rolling

direction of the respective wheel is tangentially arranged with respect to the respective rail (tangential angle of attack).

[0011] determining the real time steering angle for steering of the railway bogie, using the determined real time position and/or orientation of the railway bogie. Based on the current real time position and/or orientation of the railway bogie during operation, the real time steering angle is determined such that the railway bogie can be steered towards or is kept on the desired position/orientation with respect to the railway track. In case the set-point position and the set-point orientation corresponds to the determined position and/or orientation, a change in the current steering angle is not required. In case the set-point position and/or the set-point orientation does not correspond to the determined position and/or orientation, a change of the real time steering angle is required to guide the railway bogie back to the desired position. The real time steering angle is for example an angular value or another value, which is characteristic for the real time steering angle.

[0012] The lateral real time sensor signal measured, for example by at least one sensor, of the sensor assembly determines for example the distance of a flange of the wheel to the track. In a curve, this distance may change, which is readable or visible in the respective lateral real time sensor signal. In order to keep for example the lateral distance within a specific range to avoid contact, the railway bogie should be steered respectively. With the determined real-time steering angle, it is possible to steer the railway bogie back to the desired position or to keep the railway bogie at the desired position during operation. It is therefore possible to avoid undesired contact of the flange of the wheel with the rail. This advantageously reduces wear of the wheel and avoids noise pollution during operation.

[0013] For an advantageous determination of the position and/or orientation, the positioning algorithm may use geometry algorithms using the specific geometrical extensions of the railway bogie. The geometrical formulas may interpolate the received at least one real time sensor signal prior or during the determination of the position and/or orientation of the railway bogie.

[0014] In a variation of the disclosure, the computer-implemented method, further comprises determining by the sensor assembly at least one vertical real time sensor signal, which is characteristic for a vertical position of at least one part of the sensor assembly with respect to the railway track, and determining the real time position and/or orientation of the railway bogie with respect to the railway track, additionally uses the received vertical real time sensor signal. The sensor assembly may comprise a dedicated sensor, which is configured for measuring the vertical real time sensor signal. This sensor and the sensor for the lateral real time sensor signal may be assembled in a sensor unit forming part of the sensor assembly. The vertical real time sensor signal measures for example the vertical distance between the sensor itself and the rail. The real time vertical distance is transmitted as vertical real time sensor signal and used for the determination of the position and/or orientation of the railway bogie on the railway track by the positioning algorithm. The vertical distance may be used for the determination onto which kind of tracks the railway bogie is currently situated, this information may affect the determi-

nation of the position and/or orientation of the railway bogie. Different sections of the railway track, like junctions, switches or crossings are for example distinguishable by the vertical position of the sensor with respect to the rail. Some sections of the railway track lift the railway bogie upwards, which is visible in the vertical real time sensor signal. By further using the vertical real time sensor signal for the determination of the position and/or orientation of the railway bogie it is possible to further increase the accuracy. The vertical distance may further assist in the determination of any wear or state of wear of the railway track profile.

[0015] In a variation of the disclosure, the computer-implemented method further comprises determining by the sensor assembly a first lateral real time sensor signal, which is characteristic for a lateral position of the tread of a first wheel of the railway bogie with respect to the railway track, and a second lateral real time sensor signal, which is characteristic for a lateral position of the tread of a second wheel of the railway bogie with respect to the railway track, and determining, a real time position and/or orientation of the railway bogie with respect to the railway track, using the received first and second lateral real time sensor signals. In a variation, the first wheel and the second wheel are arranged parallel with respect to each other. For example, the first wheel is configured to contact a first rail of the railway track and the second wheel is configured to contact a second rail of the railway track. The first lateral real time sensor signal is for example characteristic for a first lateral distance between the flange of the first wheel and the first rail, and the second lateral real time sensor signal is characteristic for a second lateral distance between the flange of the second wheel and the second rail. The distance between the first and second rail is along a railway track not constant. For example in curves, at switch points or misalignment sections, the distance between the first and second rail varies, at least slightly. By taking into account the first and second lateral real time sensor signal, the geometrical center of the railway track is for example constantly updated based on the lateral real time sensor signals, which increases the accuracy of the determination of the position and/or orientation of railway bogie.

[0016] It is preferred that the computer-implemented method further comprises determining by the sensor assembly a first vertical real time sensor signal, which is characteristic for a vertical position of the first vertical sensor with respect to the railway track, and a second vertical real time sensor signal, which is characteristic for a vertical position of the second vertical sensor with respect to the railway track. Each wheel may comprise a sensor unit with a first sensor for the lateral measurement and with a second sensor for the vertical measurement.

[0017] An increase of accuracy is achievable when the computer-implemented method further comprises determining by the sensor assembly a front lateral real time sensor signal, which is characteristic for a front lateral position of the tread of a wheel of the railway bogie with respect to the railway track, and a back lateral real time sensor signal, which is characteristic for a back lateral position of the tread of a second wheel of the railway bogie with respect to the railway track. The sensor assembly may comprise a first lateral sensor arranged on the front of the respective wheel and a second sensor arranged on the back, (behind) the respective wheel, in view of the running direction of the railway bogie. In a further variation, each of the wheels of

the railway bogie comprises the front lateral sensor and the back lateral sensor. For example, a tram railway bogie with two wheels may comprise four lateral sensors. The four real time sensor signals are used for the determination of the position and/or orientation of the railway bogie. This advantageously increases the accuracy of the output.

[0018] In a further variation of the disclosure, the computer-implemented method further comprises determining by the sensor assembly a front vertical real time sensor signal, which is characteristic for a front vertical position of the sensor with respect to the railway track, and a back vertical real time sensor signal, which is characteristic for a back vertical position of the back sensor with respect to the railway track. For example, each wheel of the railway bogie may comprise a front sensor for measuring the real time vertical sensor signal in front of the respective wheel and a back sensor for measuring the real time vertical sensor signal behind the respective wheel.

[0019] The back real time sensor signals (lateral and vertical) received may in particular be used, by the positioning algorithm, for validation purposes of the front real time sensor signals.

[0020] Particularly, the sensor assembly may comprise for each wheel a front sensor unit and a back sensor unit, wherein each sensor unit comprises a first sensor for lateral measurement and a second sensor for vertical measurement, arranged next to each other. The tram railway bogie with two wheels comprises for example four sensor units, each providing a lateral real time sensor signal and a vertical real time sensor signal. The high number of sensor signals advantageously increases the accuracy for the determination of the real time position and/or orientation of the railway bogie on the railway track during its operation.

[0021] In a variation, the railway bogie may comprise four wheels, for example two wheels on two axles or four wheels with an independent suspension within the railway bogie. The sensor assembly may be configured to determine the real time sensor signals (lateral and/or vertical, front and/or back) of one wheel, of two wheels, of three wheels and/or of four wheels of such a railway bogie comprising four wheels. The steering of a railway bogie with four wheels, for example a bogie of a train, unpropelled or propelled, for example of a train wagon or of a locomotive, may be challenging in particular because steering one of the four wheels probably also affects all of the wheels of the railway bogie. It might therefore be of particular importance to monitor and control all of the lateral distances of all four wheel with respect to the respective rail.

[0022] In a variation of the present disclosure, the computer-implemented method further comprises adapting the received at least one real time sensor signal using predetermined sensor calibration data. The sensor calibration data is for example railway bogie specific or even track specific data, collected for example, during test drives along the planned operational route of the respective railway bogie. This sensor calibration data may be constantly used for adapting the received real time sensor signals. The sensor calibration data may be used to identify and categorize disturbances in the received sensor signals. In a further variation, adapting the received at least one real time sensor signal is at least supported by a trained neural network. Input data for the neural network is the raw sensor data and output data is the calibrated sensor data. The neural network is for example constantly updated based on the received sensor

data and the resulting position and/or orientation of the railway bogie on the railway track.

[0023] In a variation of the present disclosure, during the step of determining the real time position and/or orientation of the railway bogie, the positioning algorithm filters the received at least one real time sensor signal, preferably calibrated sensor signals, using at least one predefined filter parameter and determines the relevant real time sensor signal data, which are used for determining the real time position and/or orientation of the railway bogie with respect to the railway track, based on the result of the filtering. For example, a very high signal deflection may result from signal disturbances or a fault measurement. These signal deflections should be filtered out, which is performed by the positioning algorithm using the predefined filter parameters. The filter parameter may change or vary during the operation of the railway bogie. For example, the filter parameter are different on a straight section of the railway track compared to a curvy section of the railway track. The filtering creates the possibility to sort out, at least temporary, faulty sensor signals. For example, at an intersection or at a switch point, some sensor signals should not be used in the determination of the real time position and/or orientation of the railway bogie.

[0024] It is preferred that at least one predefined filter parameter depends on at least one of: a railway bogie type, a railway vehicle type, railway bogie load, a railway bogie speed and a wheel size. The filter parameters may comprise fixed parameters and/or variable parameters, which change over time.

[0025] Preferably, the step of determining the real time steering angle further uses at least one predetermined steering parameter. The steering parameters may depend on the specific railway bogie, current speed or load of the railway bogie, and/or further parameters of the railway bogie, or parameters of the railway track, like inclination angle, curvature radius and/or specific section of the current railway track. These steering parameters may be input data for the determination of the real time steering angle.

[0026] It is preferred that the determination of the real time steering angle may further comprise using a feedforward control system. It is in particular preferred that the predetermined steering parameter and/or other parameters may be used in such a feedforward control systems, in particular a predictive control system (predictive feedforward control system). In other words, the predetermined steering parameter and/or the other parameters may be used to determine a feedforward control signal as part of the real time steering angle during operation of the railway bogie. The feedforward control system improves the speed of response of the system.

[0027] In a variation of the present disclosure, the computer-implemented method further comprises the step of transmitting the determined real time steering angle to a steering actuator controller, and/or controlling a steering actuator using the steering actuator controller for steering the railway bogie around a vertical steering axis using the real time steering angle. The determined real time steering angle or a corresponding value or signal is send to the steering actuator controller, which is configured for controlling the steering actuator by means of the received real time steering angle. In a variation, the steering actuator controller uses actuator controller parameter to adjust the control command for the steering actuator. The servo drive param-

eters may depend for example on properties of the steering actuator used in the railway bogie.

[0028] In a further variation of the present disclosure, in the step of determining the real time position and/or orientation of the railway bogie, the positioning algorithm uses a neural network for determining the real time position and/or orientation of the railway bogie with respect to the railway track, wherein the at least one real time sensor signal is the input data for the neural network and the real time position and/or orientation of the railway bogie with respect to the railway track is the output data of the neural network. The neural network is preferably a trained feed-forward neural network, which provides the desired accuracy for the real time determination of the position and/or orientation of the railway bogie. Input training data of the neural network is for example lateral and vertical real time sensor signal data and the output training data for setting up (training) the neural network is the corresponding position and/or orientation of the railway bogie with respect to the railway track. The neural network is for example a feedforward neural network.

[0029] It is preferred that the neural network is trained by using the neural network for determining the real time position and/or orientation of the railway bogie with respect to the railway track, controlling the steering of the railway bogie based on the determined real time position and/or orientation of the railway bogie and receiving a feedback reward for steering of the railway bogie based on the actual position and/or orientation of the railway bogie with respect to the railway track. This reinforced learning of the neural network is for example constantly performed during operation of the railway bogie in order to constantly improve the accuracy of the determination of the position and/or orientation of the railway bogie with respect to the railway track. The feedback reward is for example the deviation of the actual position of the railway bogie from the target (set-point) position and/or target (set-point) orientation of the railway bogie. Target is to minimize this deviation over time.

[0030] Performance improvements are achievable when the neural network is trained or is additionally trained with training data obtained by operating the railway bogie on that section of the railway track for which the specific railway bogie is intended to be used. For example, the training data is collected during a test run on the specific section of the railway track. This training data is used for training of the neural network. Further, the railway bogie, which collects the training data, does not need to be the railway bogie, which operates on the railway track, the collected training data is for example transmitted to the respective railway bogie. Further, the training data used is for example selected based on the planned route of the railway bogie. This creates the possibility that the positioning algorithm, in particular the neural network, is adjusted based on the planned route of the railway bogie. Further, the parameter of the neural network may be selected based on the planned route of the railway bogie.

[0031] The sensors, which provide the sensor signals, are for example ultrasound/ultrasonic sensors, inductive sensors, laser sensors, capacitive sensors, optical sensors, radar sensors or a combination thereof.

[0032] In a variation, the computer-implemented method may further comprise to determine a rolling contact fatigue parameter of the railway track using at least one of the real time sensor signals of the sensor assembly. The sensor

signals may comprise information on the fatigue or wear of the respective railway track, which is for example read out of the respective sensor signal. The sensor signals of an ultrasound sensors may comprise such information. In a further variation, the respective fatigue or wear information is collected and stored over time and used to predict the fatigue or wear development of the respective railway track. It is therefore possible to make a prediction on the fatigue or wear development of the respective railway track using the sensor signals, feed preferably into the neural network. All railway tracks on which the respective railway bogie is operated can therefore advantageously be monitored over time, such that possible railway track cracks or failures are detected advantageously early.

[0033] According to another aspect of the present disclosure, a control device for determining a real time steering angle for a railway bogie is specified. The control device comprising a processor, which is configured to perform the following steps.

[0034] receiving, by the processor, from a sensor assembly at least one lateral real time sensor signal, which is characteristic for a lateral position of a tread of at least one wheel of the railway bogie with respect to a railway track;

[0035] determining, by the processor, by means of a positioning algorithm, a real time position and/or orientation of the railway bogie with respect to the railway track, using the received at least one lateral real time sensor signal;

[0036] determining, by the processor, the real time steering angle for steering of the railway bogie, using the determined real time position and/or orientation of the railway bogie.

[0037] In a variation of the present disclosure, the processor of the control device is further configured to receive from the sensor assembly at least one vertical real time sensor signal, which is characteristic for a vertical position of the sensor assembly with respect to the railway track, and to determine the real time position and/or orientation of the railway bogie with respect to the railway track, using additionally the received vertical real time sensor signal.

[0038] Further, the processor of the control device may be configured to receive from the sensor assembly a first lateral real time sensor signal, which is characteristic for a lateral position of the tread of a first wheel of the railway bogie with respect to the railway track, and a second lateral real time sensor signal, which is characteristic for a lateral position of the tread of a second wheel of the railway bogie with respect to the railway track and to determine a real time position and/or orientation of the railway bogie with respect to the railway track, using the received first and second lateral real time sensor signals.

[0039] In a further variation of the present disclosure, the processor of the control device is further configured to receive from the sensor assembly a first vertical real time sensor signal, which is characteristic for a vertical position of the first vertical sensor with respect to the railway track, and a second vertical real time sensor signal, which is characteristic for a vertical position of the second vertical sensor with respect to the railway track. Each wheel may comprise a sensor unit with a first sensor for the lateral measurement and with a second sensor for the vertical measurement.

[0040] In a further variation of the present disclosure, the processor of the control device is further configured to receive from the sensor assembly a front lateral real time sensor signal, which is characteristic for a front lateral position of the tread of a wheel of the railway bogie with respect to the railway track, and a back lateral real time sensor signal, which is characteristic for a back lateral position of the tread of a second wheel of the railway bogie with respect to the railway track. The sensor assembly may comprise a first lateral sensor arranged on the front of the respective wheel and a second sensor arranged on the back (behind) the respective wheel, in view of the running direction of the railway bogie. In a further variation, each of the wheels of the railway bogie comprises the front lateral sensor and the back lateral sensor. For example, a tram railway bogie with two wheels may comprise four lateral sensors. The four real time sensor signals are used for the determination of the position and/or orientation of the railway bogie. This advantageously increases the accuracy of the output.

[0041] The processor of the control device may further be configured to receive from the sensor assembly a front vertical real time sensor signal, which is characteristic for a front vertical position of the sensor with respect to the railway track, and a back vertical real time sensor signal, which is characteristic for a back vertical position of the back sensor with respect to the railway track. For example, each wheel of the railway bogie may comprise a front sensor for measuring the front real time vertical sensor signal and a back sensor for measuring the back real time vertical sensor signal.

[0042] Particularly, the sensor assembly may comprise for each wheel a front sensor unit and a back sensor unit, wherein each sensor unit comprises a first sensor for lateral measurement and a second sensor for vertical measurement. The tram railway bogie with two wheels comprises for example four sensor units, each providing a lateral real time sensor signal and a vertical real time sensor signal. The high number of sensor signals advantageously increases the accuracy for the determination of the real time position and/or orientation of the railway bogie on the railway track during its operation.

[0043] Preferably, the processor of the control device is further configured to adapt the received at least one real time sensor signal using predetermined sensor calibration data.

[0044] Advantageous results are achievable when the positioning algorithm is configured to filter the received at least one real time sensor signal using at least one predefined filter parameter, and to determine a relevant real time sensor signal data, which are used for determining the real time position and/or orientation of the railway bogie with respect to the railway track, based on the result of the filtering.

[0045] It is preferred that the at least one predefined filter parameter depends on at least one of: a railway bogie type, a railway vehicle type, railway bogie load, railway bogie speed and wheel size.

[0046] Advantageously, the processor of the control device is configured to use predetermined steering parameters for the determination of the real time steering angle.

[0047] In a variation of the present disclosure, the processor of the control device being further configured to transmit the determined real time steering angle to a steering actuator controller, which is configured to control a steering actuator

to steer the railway bogie around a vertical steering axis using the received real time steering angle.

[0048] In a further variation of the present disclosure, the processor of the control device being further configured to receive global positioning data of the railway bogie, which is used to determine the real time position and orientation of the railway bogie with respect to the railway track. The global positioning data is for example received from a GPS module, arranged in the railway bogie. The global positioning data is for example used to determine a position of the railway bogie with respect to a global reference system.

[0049] In a further variation of the present disclosure, the processor of the control device being further configured to receive Lidar data or radar data of the railway bogie, which is used to determine the real time position and orientation of the railway bogie with respect to the railway track. The Lidar data or radar data is for example received from a Lidar module and/or radar module arranged in front of the railway bogie, preferably arranged towards the railway track. The lidar data or radar data is for example used to determine a position and/or orientation of the railway bogie with respect to a local reference system or local reference points.

[0050] In a variation of the present disclosure, the processor of the control device is configured to categorize and determine sections of the railway track using the received at least one real-time sensor signals. For example, the real-time sensor signals may enable to determine on which kind of railway track the railway bogie is currently in operation. It is conceivable, that it is possible to determine if the railway bogie is currently on a straight track section, on a curvy track section, on an intersection or on a switch point section. Further, the processor of the control device may be configured to determine the real time steering angle using the determined railway track section. In a variation, the filter parameters are selected in dependence of the determined track section.

[0051] Advantageous real time application is achievable when the control device has processing speeds of 50 ms or below, preferably of 25 ms or below, even more preferably of 20 ms or below. Such processing speeds enable the desired accurate and fast determination of the real time steering angle and the desired fast and responsive steering of the railway bogie.

[0052] In a further variation of the present disclosure, the control device is configured to send a heartbeat signal to a train controller management system, the heartbeat system being configured to transmit status information of the steering system. The train controller management system has for example processing speeds of 100 ms or above. In a variation, the heartbeat signal is for example send every 100 ms.

[0053] According to a further aspect of the present disclosure, the processor of the control device is configured to perform the computer-implemented method for determining the real time steering angle as described above and hereinafter.

[0054] According to a further aspect of the present disclosure, a computer program product is specified, comprising a non-transitory computer readable medium having stored thereon computer program code configured to direct the processor of the control device to perform the method for determining the real time steering angle as described above and hereinafter.

[0055] According to a further aspect of the present disclosure, a railway bogie is specified comprising the control device as described above or hereinafter.

BRIEF DESCRIPTION OF THE DRAWINGS

[0056] The herein described disclosure will be more fully understood from the detailed description given herein below and the accompanying drawings, which should not be considered limiting to the disclosure described in the appended claims.

[0057] The drawings are showing:

[0058] FIG. 1 a perspective view of a first variation of the bogie according to the disclosure;

[0059] FIG. 2 a first perspective view of a second variation of the bogie according to the disclosure;

[0060] FIG. 3 a second perspective view of the second variation of FIG. 2;

[0061] FIG. 4 a detailed view of FIG. 3;

[0062] FIG. 5 a section view of the second variation of FIG. 2;

[0063] FIG. 6 a detailed view of FIG. 5

[0064] FIG. 7 a detailed view of a further variation of the sensor arrangement;

[0065] FIG. 8 a schematic block diagram of a control architecture of the railway bogie;

[0066] FIG. 9 a first schematic control loop for the railway bogie;

[0067] FIG. 10 a second schematic control loop for the railway bogie;

[0068] FIG. 11 a schematic block diagram of a neural network;

[0069] FIG. 12 a flow diagram illustrating schematically a plurality of steps performed by a processor for determining a real-time steering angle for the railway bogie.

DESCRIPTION OF THE EMBODIMENTS

[0070] Reference will now be made in detail to certain embodiments, examples of which are illustrated in the accompanying drawings, in which some, but not all features are shown. Indeed, embodiments disclosed herein may be embodied in many different forms and should not be construed as limited to the embodiments set forth herein; rather, these embodiments are provided so that this disclosure will satisfy applicable legal requirements. Whenever possible, like reference numbers will be used to refer to like components or parts.

[0071] FIG. 1 shows a perspective view of a first variation of the railway bogie according to the disclosure. FIG. 2 shows a first perspective view of a second variation of the railway bogie according to the disclosure. FIG. 3 a second perspective view of the second variation of FIG. 2. FIG. 4 shows a detailed view of FIG. 3. FIG. 5 shows a section view of the second variation of FIG. 2. FIG. 6 shows a detailed view of FIG. 5. FIG. 7 shows a detailed view of a further variation of the sensor arrangement. FIG. 8 shows a schematic block diagram of a control architecture of the railway bogie. FIG. 9 shows a first schematic control loop for the railway bogie. FIG. 10 shows a second schematic control loop for the railway bogie. FIG. 11 shows a schematic block diagram of a neural network. FIG. 12 shows a flow diagram illustrating schematically a plurality of steps performed by a processor for determining a real-time steering angle for the railway bogie.

[0072] As e.g. visible in FIGS. 1, 2, and 3, a railway bogie 1 comprises a base 2, which is configured to be attached to a chassis of a railway vehicle. The base 2 may be connected to a connecting part 9, which is configured to be connected to the chassis of the railway vehicle during operation of the railway bogie 1. The railway bogie 1 further comprises a frame 3 arranged rotatable with respect to the base 2 around a vertical steering axis 4. The railway bogie 1 further comprises two wheels 5, which comprise each a tread 6. The tread 6 or tread profile is the radially external portion of the wheel 5. The tread 6 comprises a contact surface or rolling surface, which is, during operation, in contact with a rail 32 of a railway track 31. The wheels 5 further comprise a wheel flange, which is conventionally configured to guide the respective wheels 5 on the rail. The wheels 5 are arranged rotatable with respect to the frame 3 around a respective wheel rotation axis 7. The wheel rotation axis 7 of the wheels 5 are arranged essentially coaxially with respect to each other and the steering axis 4 is arranged in a lateral direction Y between the two wheels 5. In another variation, the wheel rotation axis 7 may be arranged at a specific angle with respect to the lateral direction Y. In this case, the wheel rotation axis 7 are inclined with respect to the lateral direction Y. FIG. 1 further shows covers 30 arranged on the frame 3 for protection of the bogie 1 during operation. The railway bogie 1 as shown in FIG. 2 does not comprise, among other things, the connecting part 9.

[0073] The Figures further show that the frame 3 comprises a base frame 23 and a wheel frame 24. The base frame 23 is arranged mainly above the wheel frame 24. The base frame 23 is interconnected to the wheel frame 24 via a spring damping system 12 and vice versa. The spring damping system 12 comprises a damper 25 and a spring assembly 26 with a first spring 27 and a second spring 28. Struts 29 are arranged between the base frame 23 and the wheel frame 24.

[0074] FIG. 1 further shows a steering actuator 16, which is connected to the frame 3 and to the connecting part 9. Movement of the steering actuator 16 cause a rotation of the frame 3 around the steering axis 4 by a steering angle with respect to the base 2 and with respect to the connecting part 9 and also with respect to the chassis of the railway vehicle. The FIG. 1 further show a fluidic engine, which is configured to drive the steering actuator 16. The fluidic engine is for example an electrical engine, which propels a hydraulic pump for controlling the steering actuator 16.

[0075] The railway bogie 1 further comprises a sensor assembly 11, best visible in FIGS. 3 and 4. The sensor assembly 11 comprises a front sensor unit 13 arranged in front of the respective tread 6 of the wheel 5 with respect to a running direction X of the bogie 1. The sensor assembly 11 further comprises a back sensor unit 14 arranged behind the respective tread 6 of the wheel 5 with respect to the running direction X of the bogie 1. As best visible in FIG. 3, both wheels 5 of the railway bogie 1 comprise the front sensor unit 13 and the back sensor unit 14.

[0076] The front sensor unit 13 and the back sensor unit 14 are arranged on a sensor bracket 15, which is mounted pivotable with respect to the wheel 5. The sensor bracket 15 extends along the wheel 5 and holds the respective sensor units 13, 14 at a predefined position during operation of the railway bogie 1. The sensor bracket 15 is arranged pivotable with respect to the wheel around a leveling axis 10, best visible in FIG. 4. The leveling axis 10 is arranged parallel

with respect to the respective wheel rotation axis 4 and on the same vertical virtual plane as the wheel rotation axis 4. [0077] The bogie 1 further comprises leveling actuators 8, which are connected to the frame 3 and to the respective sensor bracket 15. Linear movement of the leveling actuators 8 adjusts during operation of the bogie 1 the vertical position of the front sensor unit 13 and/or of the back sensor unit 14 with respect to the rail. Movement of the frame 3, in particular movement of a swing arm 20 of the frame 3, which might affect the vertical position of at least one of the sensor units 13, 14, can be compensated by movement of the leveling actuators 8. In the variation of the disclosure as shown in the Figures, the linear movement of the leveling actuators 8 causes pivoting of the connected sensor bracket 15 around the leveling axis 10. The pivoting around the leveling axis 10 compensates a possible deflection of the swing arm 20 against a swing arm spring 22 around a pivot axis 21 during operation of the railway bogie 1, such that the vertical position of the respective sensor units 13, 14 may stay as static as possible during operation of the railway bogie 1.

[0078] FIG. 1 further show that each wheel 5 comprises an electrical engine 18 and a brake 19. The electrical engine 18 is configured to drive/accelerate, if required, during operation the respective wheel 5, and the brake 19 is configured to decelerate, if required, during operation of the railway bogie 1 the respective wheel 5. The brake 19 is a disk brake and the disk of the disk brake is arranged on the same shaft as the respective wheel 5 and the respective electrical engine 18. The wheel 5, the electrical engine 18 and the disk are partially surrounded and held by the swing arm 20, best visible in FIG. 3. A brake caliper of the brake 19 is arranged on the swing arm 20.

[0079] The Figures further indicate schematically a control device 40. The control device 40, which comprises a processor 17, is for example arranged within the railway bogie 1 or at a different position within the railway vehicle.

[0080] The FIGS. 1 and 2 further show four struts 29, which connect the base frame 23 with the wheel frame 24. The longitudinal axis of the struts 29 is arranged parallel to each other and is further arranged parallel with the running direction X in a resting state of the bogie 1. The struts 29 are connected such with the base frame 23 and the wheel frame 24 that vertical movement between these two parts is, within certain limits, enabled (damped by the spring damping assembly 12) and that movement in the running direction X is inhibited.

[0081] The Figures further shows stops 31, best visible in FIG. 2, arranged on the base frame 23, wherein some of the stops 31 limit the movement of the base frame 23 with respect to the wheel frame 24. The other stops 31 limit the maximal rotation of the frame 3 around the vertical steering axis 4.

[0082] As best visible in the FIGS. 5 and 6, the sensor assembly 11 of this embodiment comprises a plurality of sensor units 35. A front sensor unit 35 is arranged in front of the respective wheel 5 and a back sensor unit 35 is arranged behind the respective wheel. As best visible in FIG. 5, both wheels 5 comprise the two sensor units 35. FIG. 6 and FIG. 7 further show advantageously that each sensor unit 35 comprises a first sensor 36 and a second sensor 37. The first sensor 36 is configured to provide a lateral real-time sensor signal 38 and the second sensor 37 is configured to provide a vertical real time sensor signal 39. The entire sensor

assembly 11 as shown in the Figures is therefore configured to provide four lateral real-time sensor signals 38 and four vertical real-time sensor signals 39.

[0083] As best visible in the FIGS. 5, 6 and 7, the railway track 31 comprises two rails 32, which comprise at least one essentially vertical flange 33 and at least one essentially horizontal surface 34. Essentially vertical means that one extension direction of this flange extends along the vertical direction Z. Essentially horizontally means that one extension direction of this surface extends along the lateral direction Y. Rails 32 of the railway track 31 are not flat but correspond to the shape of the tread 6 of the respective wheel 5 and vice versa. In other words, the vertical flange 33 extends in the vertical direction Z and is, for example, used as a guiding surface for a flange of the tread 6 of the wheel 5. The horizontal surface 34 extends in the lateral direction Y and is for example used as a running surface for the tread 6 of the wheel 5.

[0084] FIG. 7 further shows a variation of the sensor unit 35 comprising the first sensor 36 and the second sensor 37 with a housing. The housing comprises a protective layer, which is arranged below the sensors 36, 37 for protection of the sensors 36, 37. The housing comprising the protective layer is, according to this embodiment, molded around the sensors 36, 37. The material of the housing is a nonconductive material, for example an epoxy-based resin, and is therefore at least partially transparent for the sensor measurements.

[0085] FIG. 8 illustrates schematically a block diagram of a real time control architecture of the railway bogie 1. The control architecture comprises the railway bogie 1, the control device 40 with its processor 17 and a steering actuator controller 41. These three parts are illustrated via a respective block. The railway bogie 1, in particular the sensor assembly 11 is configured to provide the real time sensor signals 38, 39. The control device 40 is configured to receive the real time sensor signals 38, 39 and to determine a real time steering angle 58. The control device is further configured to transmit the real time steering angle 58, or a respective signal, to the steering actuator controller 41, which is configured to translate the real time steering angle 58 to a corresponding steering actuator command 59, which is transmitted to the steering actuator 16 for steering of the railway bogie 1. The control device 40 comprises an operation system 42, which is configured to operate the control device 40, in particular by using at least one controller device parameter 51. The steering actuator controller 41 also comprises an operation system 43, configured to operate the steering actuator controller 41, in particular by using additionally at least one steering controller parameter 56. The parameters 51, 54 may be stored in the respective control device 40, 41.

[0086] The operation system 42 of the control device 40 comprises a main logic 44, which is configured to determine the real time steering angle 58. The main logic 44 is schematically illustrated by a sensor data block 46, a positioning algorithm block 47 and a steering control block 49. The sensor data block 46, is configured to receive real time sensor data 38, 39 from the sensor assembly 11. The sensor data 38, 39 may be adapted or processed using sensor calibration parameters 53, which are for example stored on the control device 40 and accessed by the processor 17. The positioning algorithm block 47, is configured to determine the position and/or orientation of the railway bogie 1 with

respect to the railway track 31 using the received real time sensor data 38, 39. The positioning algorithm 47 may use filtering to process the received sensor data 38, 39, the filtering is schematically shown via a filtering block 48. The filtering may be performed using filter parameter 54, which are for example stored on the control device 40 and accessed by the processor 17. The main logic 44 further comprises a steering control block 49, which is configured to determine the real time steering angle 58 or a respective value/signal using the determined position and/or orientation of the railway bogie 1 with respect to the railway track 31. The steering control block 49 may use a PID controller 50 to constantly determine the real time steering angle 58 during operation. Additionally, the steering control block 49 may use steering parameter 55, which are for example stored on the control device 40 and accessed by the processor 17. FIG. 8 further shows a safety application block 45, which uses safety parameters. The safety application block 45 illustrates schematically that different safety measures of the railway bogie 1 are in place in order to prevent different kind of failures. The safety applications 45 may use different safety parameters 52, which are for example stored on the control device and accessed by the processor 17, to control the railway bogie 1. FIG. 8 further shows schematically a gps module 70 and a radar/lidar module 71. The gps module 70 is configured to collect gps data during operation of the railway bogie 1. The radar/lidar module 71 is configured to collect radar data and/or lidar data during operation of the railway bogie 1. This data is in a variation also transmitted to the control device 40 for determining by means of the positioning algorithm 47 the real time position and/or real time orientation of the railway bogie 1 with respect to the railway track. The gps module 70 and/or the radar/lidar module 71 may be arranged at the railway vehicle, which comprises the railway bogie 1.

[0087] FIG. 9 illustrates schematically a first variation of a control loop, which is configured for determining constantly during operation of the railway bogie 1 the realtime steering angle 58. The set value 57 of this control loop is that the mechanical center of the railway bogie 1 lies exactly above the center of the railway track 31, the deviation should be zero. The mechanical center of the railway bogie 1 is for example the point at the half distance between the two wheels 5 of the railway bogie 1. The center of the railway track 31 is for example the point at the half distance between the two rails of the railway track 31. The set value is that the lateral distance between the mechanical center of the railway bogie 1 and the center of the railway track 31 is zero. The feedback value or recirculation value is or are the real time sensor signals 38, 39. The control error 66 of the control loop is the difference between the set value 57 and the actual real time position and/or orientation of the railway bogie 1 with respect to the railway track 31 determined using the sensor signals 38, 39. A correction input 67 is determined, for example by the processor 17 of the control device 40, using the control error 66. The correction input 67 is for example determined using a PID controller. The correction input 67 is for example the real time steering angle 58. The real time steering angle 58 is used for a respective steering of the railway bogie 1. Without disturbances 68, the railway bogie 1 may be steered such that the set point position corresponds to the actual position. Disturbances 68, which might result from a change in the railway track 31, disturbances in the steering actuator 16 or in its steering actuator controller 41

result in a deviation of the actual position with respect to the set point position. This deviation 69 is determined using the sensor signals 38, 39 and fed back.

[0088] FIG. 10 illustrates schematically a second variation of a control loop, which is configured for determining constantly during operation of the railway bogie 1 the real-time steering angle 58. The control loop of the second variation differs from the control loop of the first variation in that it further comprises a feedforward controller 72, which may increase the control speed of the control loop. The feedforward controller 72 may receive steering parameter data 55 and the set value. The feedforward controller 72 may determine control input for the steering actuator controller 41 for directly controlling the actuator controller 41. The feedforward controller 72 increases the control speed and may be used alternatively or additionally to the control loop of the first variation. The feedforward controller 72 may use a neural network for determining the output data. The feedforward controller 72 is for example implemented in the control device 40 and/or the processor 17.

[0089] FIG. 11 illustrates schematically a block diagram of a neural network 60, which might be used by the positioning algorithm 47 for the determination of the real time position and/or orientation of the railway bogie 1 with respect to the railway track 31. The neural network 60 comprises an input layer 63, a hidden layer 64 and an output layer 65. Input data 61, in particular at least one lateral real time sensor signal 38 and at least one vertical real time sensor signal 39 is, is fed into the neural network 60. The hidden layer 64 processes the received data and the neural network 60 produces as output data 62 the real time position and/or orientation of the railway bogie 1 with respect to the railway track 31.

[0090] FIG. 12 shows a flow diagram illustrating schematically a sequence of steps performed by the processor 17 of the control device 40 for determining a realtime steering angle 58 for the railway bogie 1 on a current section of the railway track. The sequence of steps, implemented for example as a computer-implemented method, is for example performed by the control device 40 as shown in FIG. 8.

[0091] In step S1, the processor 17 of the control device 40, receives from the sensor assembly 11 at least one lateral real time sensor signal 38, which is characteristic for the lateral position of the tread 6 of the at least one wheel 5 of the railway bogie 1 with respect to a railway track 31. The sensor assembly 11 comprising the plurality of sensor units 35 with the first sensor 36 and the second sensor 37 sends or transmits the real time sensor signals to the control device 40.

[0092] In the optional step S4, the processor 17 of the control device 40, adapts the received at least one real time sensor signal using the predetermined sensor calibration data 53. In this step, which might also be performed by the sensor assembly 11 itself, the sensor signals 38, 39 are adapted, for example smoothed or averaged, by using the predetermined sensor calibration data 53.

[0093] In step S2, the processor 17 of the control device 40 determines by means of the positioning algorithm 47 the real time position and/or orientation of the railway bogie 1 with respect to the railway track 31, using the received at least one lateral real time sensor signal. The current or actual (real-time) position and/or orientation of the railway bogie 1 is determined based on the received real time sensor signals,

the positioning algorithm 47 may use geometry algorithms, interpolation algorithms, neuronal networks and/or a combination thereof.

[0094] In step S3, the processor 17 of the control device 40 determines the real time steering angle 58 for steering of the railway bogie 1, using the determined real time position and/or orientation of the railway bogie 1 with respect to the railway track 31. The real time steering angle 58 is for example determined additionally using steering parameters 55, like velocity, additional track information etc. The steering angle 58 is for example an angular value or a signal, which corresponds to an angular value. The real time steering angle 58 is for example the angle between the current (tangential) main extension direction of the rail 32 and the running direction of the respective wheel 5. The real time steering angle 58 may also be the angle between the running direction of the railway bogie 1 or the wheel 5 and a chassis or connecting part 9 of a railway vehicle. In both variations, a change in the real time steering angle 58 results in a lateral displacement of the railway bogie 1 with respect to the railway track during operation. A target value of the real time steering angle 58 is for example the curvature angle of the railway track 31.

[0095] In step S5, the processor of the control device 40 transmits the determined real time steering angle 58 to the steering actuator controller 41. The steering actuator controller 41 is configured to translate the received real time steering angle 58 to the desired control instructions for the steering actuator 16, which is configured to swivel or rotate the railway bogie 1 around the vertical steering axis 4.

[0096] In step S6, the steering actuator 16 is controlled by the steering actuator controller 41 for steering the railway bogie 1 around the vertical steering axis 4 using the real time steering angle 58. The steering actuator 16 is constantly in operation to swivel/rotate the railway bogie 1 around the vertical steering axis 4 such that the actual position and/or orientation of the railway bogie 1 is as aligned as possible with the target (set-point) position and/or orientation during operation of the railway bogie 1.

[0097] At least a portion of the steps is preferably performed constantly to keep the railway bogie 1 perfectly aligned on the railway track 31. Disturbances 68, which might affect the control loop, are taken into account such that a smooth running of the railway bogie 1 on the railway track 31 is achieved.

[0098] Rather, the words used in the specification are words of description rather than limitation, and it is understood that various changes may be made without departing from the Spirit and scope of the disclosure.

LIST OF DESIGNATIONS

1 Railway Bogie	25 Damper
2 Base	26 Spring assembly
3 Frame	27 First spring
4 Steering axis	28 Second Spring
5 Wheel	29 Strut
6 Tread	30 Cover
7 Wheel rotation axis	31 Railway track
8 Leveling actuator	32 Rail
9 Connecting part	33 Vertical flange
10 Leveling axis	34 Horizontal flange
11 Sensor assembly	35 Sensor unit
12 Spring damping system	36 First sensor
13 Front sensor	37 Second sensor

-continued

LIST OF DESIGNATIONS

14 Back sensor	38 Lateral real time sensor signal
15 Sensor bracket	39 Vertical real time sensor signal
16 Steering actuator	40 Control device
17 Processor	41 Steering actuator controller
18 Electrical engine	
19 Brake	
20 Swing arm	
21 pivot axis	42 Operation system control device
22 Swing arm spring	43 Operation system steering actuator controller
23 Base frame	64 Hidden layer
24 Wheel Frame	65 Output layer
44 Main logic	66 Control error
45 Safety Application block	67 Correction input
46 Sensor data block	68 Disturbances
47 Position algorithm block	69 Deviation
48 Filtering block	70 GPS module
49 Steering control block	71 Radar/Lidar module
50 PID controller	72 feedforward controller
51 Control device parameter	
52 Safety parameter	
53 Sensor calibration data	X running direction
54 Filter parameter	Y lateral direction
55 Steering parameter	Z vertical direction
56 Steering actuator controller-parameter	S1 Receiving
57 Set value (Mechanical center)	S2 Determining position
58 Real time steering angle	S3 Determining steering angle
59 Steering actuator control commands	S4 adapting the received real time sensor signal(s)
60 Neural network	S5 transmitting steering angle
61 Input data	
62 Output data	S6 controlling steering actuator
63 Input layer	

1. A computer-implemented method for determining a real time steering angle for a railway bogie, the method comprising:

- determining by a sensor assembly at least one lateral real time sensor signal, which is characteristic for a lateral position of a tread of at least one wheel of the railway bogie with respect to a railway track;
- determining, by means of a positioning algorithm, a real time position and/or orientation of the railway bogie with respect to the railway track, using the determined at least one lateral real time sensor signal; and
- determining the real time steering angle for steering of the railway bogie, using the determined real time position and/or orientation of the railway bogie.

2. The computer-implemented method according to claim 1, further comprising:

- determining by the sensor assembly at least one vertical real time sensor signal, which is characteristic for a vertical position of the sensor assembly with respect to the railway track, and
- determining the real time position and/or orientation of the railway bogie with respect to the railway track, further using the received vertical real time sensor signal.

3. The computer-implemented method according to claim 1, further comprising:

- determining by the sensor assembly a first lateral real time sensor signal, which is characteristic for a lateral position of the tread of a first wheel of the railway bogie with respect to the railway track, and a second lateral real time sensor signal, which is characteristic for a

- lateral position of the tread of a second wheel of the railway bogie with respect to the railway track;
- b. determining, the real time position and/or orientation of the railway bogie with respect to the railway track, using the received first and second lateral real time sensor signals.
4. The computer-implemented method according to claim 1, further comprising:
- a. determining by the sensor assembly of at least one wheel of the railway bogie a front lateral real time sensor signal, which is characteristic for a lateral position of the tread of the wheel, a front vertical real time sensor signal, which is characteristic for a vertical position of the sensor assembly, using a front sensor unit of the sensor assembly of the wheel and/or determining by the sensor assembly of the at least one wheel of the railway bogie a back lateral real time sensor signal, which is characteristic for a lateral position of the tread of the wheel, a back vertical real time sensor signal, which is characteristic for a vertical position of the sensor assembly, using a back sensor unit of the sensor assembly of the wheel;
 - b. determining, the real time position and/or orientation of the railway bogie with respect to the railway track, using the received front and back real time sensor signals.
5. The computer-implemented method according to claim 1, further comprising:
- a. adapting the received at least one real time sensor signal using predetermined sensor calibration data.
6. The computer-implemented method according to claim 1, wherein in the step of determining the real time position and/or orientation of the railway bogie, the positioning algorithm filters the received at least one real time sensor signal using at least one predefined filter parameter and determines the relevant real time sensor signal data, which are used for determining the real time position and/or orientation of the railway bogie with respect to the railway track, based on the result of the filtering.
7. The computer-implemented method according to claim 6, wherein the at least one predefined filter parameter depends on at least one of: a railway bogie type, a railway vehicle type, a railway bogie load, a railway bogie speed and a wheel size.
8. The computer-implemented method according to claim 1, wherein the step of determining the real time steering angle further uses predetermined steering parameters.
9. The computer-implemented method according to claim 1, further comprising:
- a. transmitting the determined real time steering angle to a steering actuator controller, and
 - b. controlling, by the steering actuator controller, a steering actuator for steering the railway bogie around a vertical steering axis using the real time steering angle.
10. The computer-implemented method according to claim 1, wherein in the step of determining the real time position and/or orientation of the railway bogie, the positioning algorithm uses a neural network for determining the real time position and/or orientation of the railway bogie with respect to the railway track, wherein the at least one real time sensor signal is the input data for the neural network and the real time position and/or orientation of the railway bogie with respect to the railway track is the output data of the neural network.

11. The computer-implemented method according to claim 10, wherein the neural network is trained by:

- a. using the neural network for determining the real time position and/or orientation of the railway bogie with respect to the railway track, controlling the steering of the railway bogie based on the determined real time position and/or orientation of the railway bogie and receiving a feedback reward for steering of the railway bogie based on the actual position and/or orientation of the railway bogie with respect to the railway track for training of the neural network.

12. The computer-implemented method according to claim 10, wherein the neural network is trained with training data obtained by operating the railway bogie on that section of the railway track for which the specific railway bogie is intended to be used.

13. The computer-implemented method according to claim 1, further comprising to determine a rolling contact fatigue parameter of the railway track using at least one of the real time sensor signals of the sensor assembly.

14. A control device for determining a real time steering angle for a railway bogie, the control device comprising a processor, which is configured to perform the following steps:

- a. receiving, by the processor, from a sensor assembly at least one lateral real time sensor signal, which is characteristic for a lateral position of a tread of at least one wheel of the railway bogie with respect to a railway track;
- b. determining, by the processor, by means of a positioning algorithm, a real time position and/or orientation of the railway bogie with respect to the railway track, using the received at least one lateral real time sensor signal; and
- c. determining, by the processor, the real time steering angle for steering of the railway bogie, using the determined real time position and/or orientation of the railway bogie.

15. The control device according to claim 14, wherein the processor is further configured to:

- a. receiving, by the processor, from the sensor assembly at least one vertical real time sensor signal, which is characteristic for a vertical position of the sensor assembly with respect to the railway track, and
- b. determining, by the processor, the real time position and/or orientation of the railway bogie with respect to the railway track, using additionally the received vertical real time sensor signal.

16. The control device according to claim 14, wherein the processor is further configured to:

- a. receiving, by the processor, from the sensor assembly a first lateral real time sensor signal, which is characteristic for a lateral position of the tread of a first wheel of the railway bogie with respect to the railway track, and a second lateral real time sensor signal, which is characteristic for a lateral position of the tread of a second wheel of the railway bogie with respect to the railway track;
- b. determining, by the processor, the real time position and/or orientation of the railway bogie with respect to the railway track, using the received first and second lateral real time sensor signals.

17. The control device according to claim **14**, wherein the processor is further configured to adapt the received at least one real time sensor signal using predetermined sensor calibration data.

18. The control device according to claim **14**, wherein the processor is further configured, by means of the positioning algorithm, to filter the received at least one real time sensor signal using at least one predefined filter parameter, and to determine relevant real time sensor signal data, which are used for determining the real time position and/or orientation of the railway bogie with respect to the railway track, based on the result of the filtering.

19. The control device according to claim **18**, wherein the at least one predefined filter parameter depends on at least one of: a railway bogie type, a railway vehicle type, a railway bogie load, a railway bogie speed and a wheel size.

20. The control device according to claim **14**, wherein the processor is configured to further use predetermined steering parameters for the determination of the real time steering angle.

21. The control device according to claim **14**, the processor being further configured to transmit the determined real time steering angle to a steering actuator controller, which is configured to control a steering actuator to steer the railway bogie around a vertical steering axis using the received real time steering angle.

22. The control device according to claim **14**, wherein the processor is configured to receive and process the at least one real time sensor signals from an ultrasound/ultrasonic sensor, an inductive sensor, a laser sensor, a capacitive sensor, an optical sensor and/or a radar sensor of the sensor assembly.

23. The control device according to claim **14**, wherein the processor is further configured to determine a rolling contact fatigue parameter of the railway track using at least one of the real time sensor signals received from the sensor assembly.

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